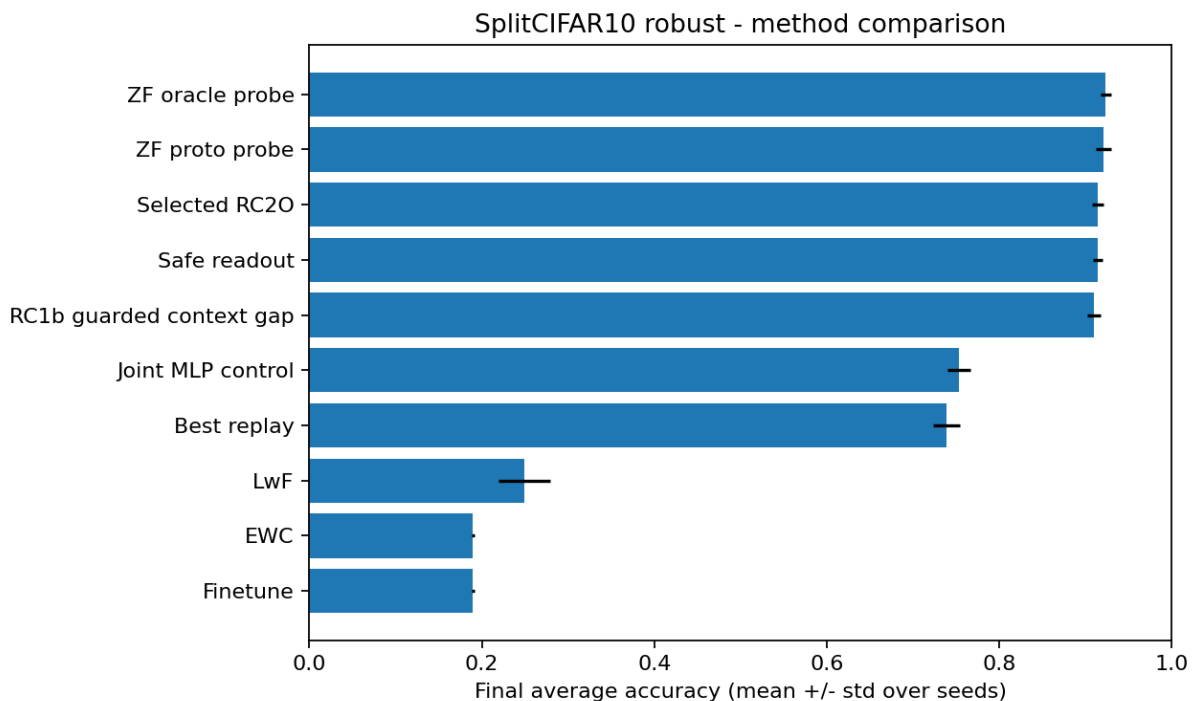


MMALS / RC2O-v2.2c - SplitCIFAR10 Robust Analysis

5-seed robust run over frozen A1 visual features; external continual-learning bridge, not end-to-end raw-pixel continual representation learning.

Verdict. The robust run confirms the evidence result. Selected RC2O reaches **0.9140** +/- 0.0064 final average accuracy over 5 seeds, against **0.7386** +/- 0.0156 for the best validation-selected replay baseline. The paired delta is **+0.1754** with 95% CI [0.1609, 0.1898]. Min-task, min-class and zero-recall checks remain healthy.

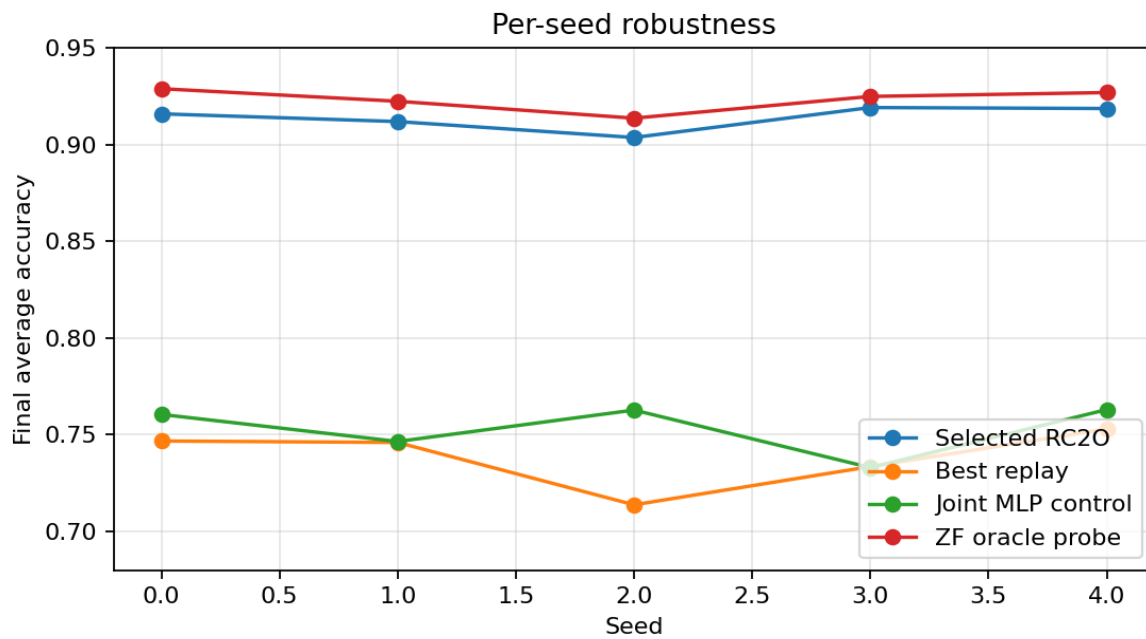
Field	Value
Run mode	robust
Dataset	SplitCIFAR10 class-incremental, 5 tasks x 2 classes
Seeds	[0, 1, 2, 3, 4]
Epochs/task	8
Memory/class	256
Input representation	small_cnn_feature
Feature dim	256
Device	cuda
Noise config	label=0.1, input=0.05, context=0.2



1. Core quantitative result

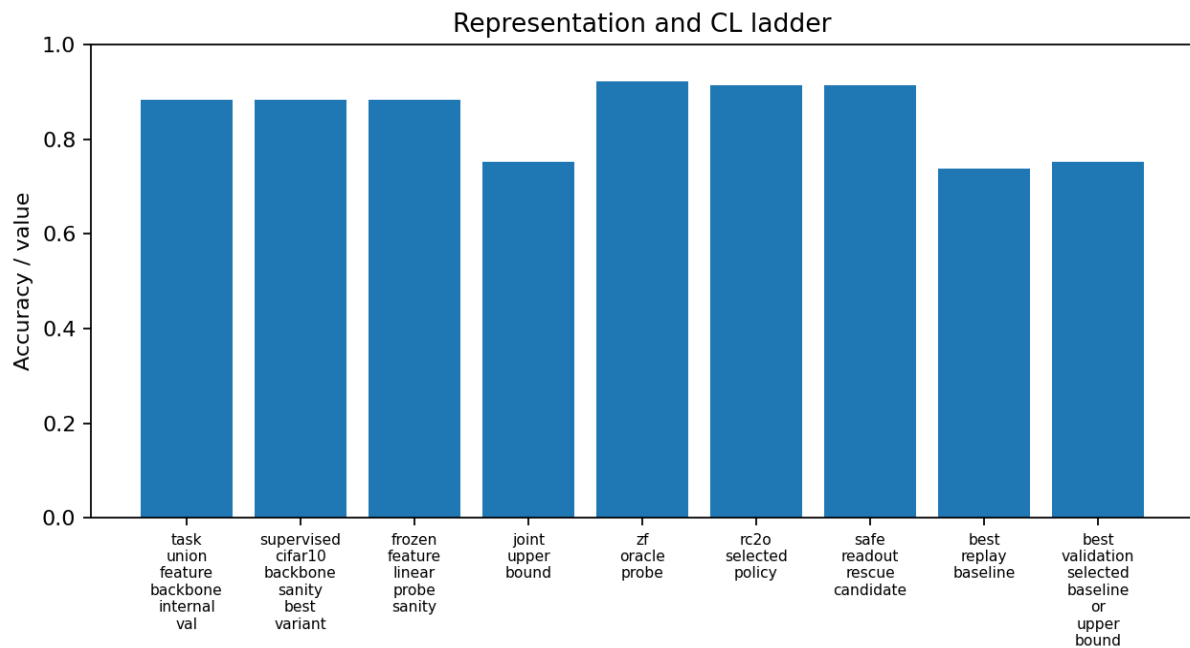
Method	Final avg acc	Std	Min-task	Forgetting	Min-class	Class 2	Class 5
ZF oracle probe	0.9234	0.0059	0.8678	NA	0.8270	0.8855	0.8950
ZF proto probe	0.9206	0.0086	0.8642	NA	0.8140	0.9145	0.8715
Selected RC2O	0.9140	0.0064	0.8535	0.0242	0.8410	0.8530	0.8850
Safe readout	0.9139	0.0054	0.8530	0.0239	0.8430	0.8560	0.8925
RC1b guarded context gap	0.9097	0.0081	0.8470	0.0298	0.8285	0.8450	0.8625
Joint MLP control	0.7531	0.0132	0.6245	0.0000	0.5795	0.6790	0.6335
Best replay	0.7386	0.0156	0.6020	0.1921	0.5665	0.6305	0.6745
LwF	0.2498	0.0299	0.0380	0.0039	0.0005	0.0905	0.0770
EWC	0.1899	0.0016	0.0000	0.9346	0.0000	0.0000	0.0000
Finetune	0.1897	0.0017	0.0000	0.9306	0.0000	0.0000	0.0000

Interpretation: RC2O is within about 0.0095 of the diagnostic ZF oracle probe, while beating replay by about 0.1754 average accuracy points. The residual oracle gap is small but non-zero, so there is still headroom for selector/readout/context refinement.



2. Representation ladder and external-bridge gate

Variant	Supervised val	Supervised test	Probe val	Probe test	Params
smallcnn_v21_reference	0.7445	0.7456	0.7510	0.7502	129,290
smallcnn_a1_aug_norm	0.8844	0.8797	0.8841	0.8803	625,482
resnet18_imagenet_frozen	0.8682	0.8598	0.8690	0.8558	11,181,642



Gate	Value	Floor/Ceiling	Pass
a1 supervised backbone pass	0.8844	≥ 0.6500	True
a1 frozen linear probe pass	0.8841	≥ 0.5500	True
joint upper bound representation floor	0.7531	≥ 0.5500	True
selected rc2o competitive with replay	0.9140	≥ 0.7286	True
selected min task nonzero	0.8535	≥ 0.0500	True
selected forgetting not explosive	0.0242	≤ 0.2000	True
selected min class nonzero	0.8410	≥ 0.0500	True
selected zero recall classes gate	zero_recall_classes=[]	≤ 0.0000	True
specialization components materialized	4.0000	≥ 4.0000	True
specialization aligned to selected policy	1.0000	≥ 1.0000	True

3. Robustness, collapse, and specialization audit

The readout-collapse audit is clean for selected RC2O, safe readout, RC1b, replay, and joint control: zero-recall classes are absent for these main methods. LwF still collapses on most seeds. The specialization audit materializes all four requested components and is aligned to the selected policy candidate.

Method	Effective classes	Min recall	Collapse warn rate
Best replay	9.8486	0.5665	0.0000
Joint MLP control	9.9237	0.5795	0.0000
LwF	2.6720	0.0005	0.8000
Safe readout	9.9903	0.8430	0.0000
RC1b guarded context gap	9.9865	0.8285	0.0000
Selected RC2O	9.9873	0.8410	0.0000

Component	OK seeds	Rows
host_ablation	5	150
host_probe	5	125
prototype_separation	5	50
route_context_and_confusion	5	50

Diagnostic audits remain diagnostic only: host ablation/probes and LSA show evidence of functional specialization, but they should not be described as full causal proof of biological-style host specialization. The robust paper claim should remain: selected-policy specialization audit materialized and aligned.

4. Baseline hardening and selector behavior

The robust package contains 120 baseline-hardening candidate traces across five seeds. Baselines are validation-selected, not arbitrary defaults. Replay remains the strongest deployable baseline family, while EWC, finetune, MoE and PNN collapse in this feature-based SplitCIFAR10 setting.

Baseline	n selected	Val score	Val final acc	Val min-task	Val forgetting
balanced_replay	5	0.7238	0.6973	0.5577	0.1171
experience_replay	5	0.7238	0.6973	0.5577	0.1171
lwf	5	0.2289	0.2285	0.0190	0.0063
sparse_moe	5	0.0066	0.1759	0.0000	0.6769
pnn	5	0.0062	0.1746	0.0000	0.6737
noisy_topk_moe	5	0.0059	0.1751	0.0000	0.6765
ewc	5	0.0051	0.1753	0.0000	0.6806
hard_top1_moe	5	0.0050	0.1729	0.0000	0.6715
dense_moe	5	0.0043	0.1737	0.0000	0.6777
finetune	5	0.0040	0.1733	0.0000	0.6771
joint_upper_bound	5	NA	NA	NA	NA

Selected policy family	Count	Mean proxy acc	Mean proxy score
context_gap_safe_class_balanced_readout_calibrated	16	0.8908	1.4264
context_temp_blend_selected_head_guard_035	6	0.8830	1.4985
proto_global_no_repair	3	0.8976	1.4926

Weak-seed audit: 3 seeds are marked for localized inspection relative to the older RC1b/true-guard comparison, mainly because guard variants sometimes slightly outperform RC1b on proxy/min-class details. This does not overturn the robust selected-RC2O result; it indicates a useful next diagnostic on selector regret and local policy choice.

5. Claim boundary and next step

Claim now supported. RC2O-v2.2c provides a robust 5-seed SplitCIFAR10 external-bridge result over a frozen, supervised A1 feature representation: selected RC2O substantially outperforms validation-selected replay and keeps low forgetting, healthy min-task accuracy, healthy min-class recall, and no zero-recall classes.

Claim not yet supported. This is not end-to-end raw-pixel continual representation learning, not broad multi-dataset robustness, and not a SOTA CIFAR10 claim. The joint MLP row should be called a joint/same-readout control, not a theoretical upper bound, because selected RC2O exceeds it.

Next experimental move. Freeze this as the SplitCIFAR10 robust bridge. Then run SplitCIFAR100 with the same reporting discipline, and separately design a Level-B slow-plasticity perceptual-host experiment where A1 can adapt under stability constraints.

Primary derived artifacts in this addendum package:

- data/robust_analysis_summary.json
- data/robust_key_methods.csv
- data/robust_per_seed_core_comparison.csv
- data/robust_representation_ladder.csv
- data/robust_baseline_hardenig_summary.csv
- data/robust_readout_collapse_focus.csv
- data/robust_specialization_status.csv